FERBIN RICHARD

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EDUCATION

Northeastern University, Boston, MA **Sep 2025 - April 2027**
Master of Science in Robotics

Vellore Institute of Technology, Chennai, India **Sep 2021 - April 2025**
Bachelor of Technology in Mechatronics, Robotics and Automation Engineering

TECHNICAL SKILLS

Programming Languages: Python, C++, Embedded C, Bash, CMake
Robotics Frameworks: ROS 2 (Nav2, MoveIt2), SLAM (GTSAM iSAM2 factor graphs, RTAB-Map, SLAM Toolbox), MAVROS/MAVLink/pymavlink, Drone & ROV autopilots (ArduPilot, ArduSub, PX4)
Computer Vision & AI: OpenCV, YOLO (v7 to v11), PyTorch, TensorFlow, Keras, Open3D, sensor fusion
Simulation & CAD: Gazebo (Harmonic, Classic & Ignition), NVIDIA Isaac Sim, SolidWorks, Fusion 360
Hardware & Tools: Pixhawk, Jetson Orin Nano, Raspberry Pi 5/4, ESP32-S3, Ping360 sonar, RealSense D455, I2C & serial bring-up, Docker, Git, Linux

PROFESSIONAL EXPERIENCE

Robotics Engineer (Co-op), Archenaux Robotics, Boston, MA **June 2026 - September 2026**

- Primary engineer for the vehicle's sensor stack and operator interface: Ping360 scanning sonar, three cameras, VectorNav VN-100 IMU, Bar30 depth, lighting and the PyQt operator UI. Diagnosed a sonar driver returning a fake wall at 0.3 m on every beam while the real wall sat at 4 to 5 m, invisible downstream because message rates looked valid, and fixed it with a regression test; raised the serial link to 2 Mbaud, taking a 360-degree sweep from 23.3 s to 5.9 s; and characterized the IMU the estimate depends on over 321k samples (Allan variance, Kalibr noise parameters, gyro ARW 0.26 deg/sqrt-hr).
- Sole author of the containerized runtime that ships to the real ROV: MAVLink bridge on the companion computer, joystick piloting, mechanism control, and video, as one multi-arch Docker image deployable on a blank laptop with a single script.
- Trained an underwater object detector with no real training data by generating a seeded procedural world that emits ground-truth labels via 3D-to-2D projection, reaching 0.969 mAP@50 across two diagnosed fine-tuning rounds and cutting clutter false positives from 54% to 0% (YOLO11, PyTorch).
- Brought up the vehicle's sensors, fixing a dead depth sensor traced to a BARO_PROBE_EXT I2C bitmask, an RC-override system-ID mismatch blocking all thruster commands, and tether rail sag misread as thermal throttling; authored the power-tree analysis behind the companion-computer swap.
- Owned the shared three-container simulation core (ArduSub SITL, Gazebo Harmonic, QGroundControl) the other engineers build on, and ported the vehicle to NVIDIA Isaac Sim as a second physics backend with a Gerstner-wave ocean and hydrodynamics, leading the co-op engineering team reporting to the CEO across five repositories.

Vice-President (Robotics Team), Vellore Institute of Technology, India **Oct 2022 - Aug 2024**

- Co-led a team of 80 to design and manufacture FPV drones, autonomous drones, battle bots, and AMRs.
- Organized workshops and competitions to build robotics talent, mentoring students in ROS2, Gazebo, and OpenCV, and developed algorithms for path planning, navigation, and sensor fusion.

Robotics Intern, Mafkin Robotics, Chennai, India **Jan 2024 - June 2024**

- Engineered an autonomous ship hull cleaning robot to address biofouling, which increases drag by up to 40% and drives higher fuel costs and carbon emissions.
- Implemented real-time 3D SLAM in Gazebo using RTAB-Map, enabling precise mapping and navigation for underwater environments, and collaborated on a cavitation jet based cleaning system.

Research Intern, Universiti Teknologi Petronas, Perak, Malaysia **Sep 2023 - Dec 2023**

- Researched autonomous drone and AMR technologies for surveillance and navigation, applying ROS2 for system integration and control.
- Implemented and tested SLAM and navigation algorithms in Gazebo, and integrated multi-sensor perception using Intel RealSense D455i depth cameras, RPLIDAR, wheel encoders, and IMUs.

PROJECTS

Autonomous Vehicle Perception-to-Control Stack, github.com/FERBIN12/av-perception-planning-stack **2026**

- Twelve-module driving stack in pure Python: camera and lidar perception, sensor fusion, localization, tracking, prediction, planning, control, and a fail-closed decision contract that refuses to drive on missing, stale or wrong-frame data; 227 tests, NumPy the only dependency.

Visual SLAM and VIO Estimation Stack, github.com/FERBIN12/visual-slam-vio-ros2 **2026**

- Visual-inertial estimation from scratch: camera calibration, two-view geometry and PnP, robust bundle adjustment, IMU preintegration with Monte Carlo-validated covariance, camera-IMU extrinsic and time-offset estimation, and sliding-window optimization with Schur-complement marginalization, evaluated by ATE/RPE with SE(3)/Sim(3) alignment in 25 seeded experiments with known ground truth.

Gazebo to Isaac Sim Migration Harness, github.com/FERBIN12/gazebo-to-isaac-migration **2026**

- Migrated a ROS 2 diff-drive rover from Gazebo Harmonic to Isaac Sim with a harness that quantifies where two physics engines diverge rather than asserting they match; its per-field tolerances failed on spin, traced to the importer mapping joint damping into the PhysX drive gain, and it caught a permuted inertia axis and four planted model defects.

ROS 2 C++ Mobile Robot Stack, github.com/FERBIN12/self_driving **Mar 2025 - Jul 2025**

- Eleven-package ROS 2 stack in C++: A* and Dijkstra global planners on an occupancy grid using TRANSIENT_LOCAL map QoS and tf2, a Kalman filter and odometry motion model for localization, mapping with known poses, differential-drive controllers, and a ros2_control hardware interface driving a serial microcontroller.

AWARDS

- **Won Track 2, The Open Accelerator healthcare hackathon**, hosted by Red Hat with IBM Ventures and the Mass AI Hub. **2026**
- **First prize, maze-solving robot competition**, hosted by IIT Hyderabad. **Feb 2023**

PUBLICATIONS

- Automated Detection and Classification of Rice Leaf Diseases Using YOLOv8, **IET** **Jan 2025**
- Efficient Real-Time Okra Stage Identification Using YOLOv8, **IEEE** **Sep 2024**
- Power Transmission Line Inspection Using Unmanned Aerial Vehicle: A Review, **IEEE** **Feb 2024**